

Chapter 6

Development of an MVA-Based High Level Trigger

6.1 Limitations and Benefits of Current $H \rightarrow \gamma\gamma$ Trigger

As discussed in section 3.3, the current $H \rightarrow \gamma\gamma$ HLT uses hard cuts on kinematic, isolation, and shower shape variables to select events. While cuts on some of these variables – such as R_9 and H/E – serve to remove low quality photon candidates, others reduce the phase space of the trigger while not necessarily increasing the quality of the analysis. Principally, the current HLT makes rather tight cuts on the p_T of the two photon candidates. These cuts significantly reduce the trigger rate, as many background processes produce showers with low p_T . Many signal events, however, have at least one candidate with low p_T . It will be shown in section 6.3 that these p_T cuts significantly decrease the acceptance of simulated Monte Carlo signal events. Since many future analyses will be searching for either very rare processes or physics beyond our current understanding of the standard model, constraining the phase space of available information may affect the collaboration's ability to probe novel processes.

The relative simplicity of the standard HLT, however, provides multiple benefits. Firstly, the numerical cuts can be closely examined and understood very easily. Though the cuts have been relatively stable in time, they may adjusted and reexamined with relative ease. The effects of cuts on individual variables can be quantified, which allows maximum transparency into how a decision is made on an event-by-event basis. Secondly, a decision for each event can be made

① important to measure kinematic distributions of the Higgs. (For example) p_T and dist. in associated production.

very rapidly. As discussed in section ??, minimizing computation time for each event is vital given the huge number of triggers deployed at the trigger farm. Since all of the quantities used by the current HLT are already being saved for each event, there is very little additional computation needed to arrive at a triggering decision. Any proposed changes to the $H \rightarrow \gamma\gamma$ trigger must have minimal impact on decision speed given the stringent restrictions on computation time.

6.2 Development of a Boosted Decision Tree for Photon Candidate Classification

The HLT is a binary classification problem, where the two classes are signal events to keep and background events to discard. While the HLT is a per-event selection, $H \rightarrow \gamma\gamma$ events contain two photon candidates. As described in 3.3, the current HLT first makes requirements on individual photons, then applies a final filter that considers both photons together. Our MVA-HLT uses a similar philosophy; a binary classifier can be used to calculate an identification score for each photon, then combined cuts can be made on these scores for two photon candidates. The process of creating the MVA-HLT, then, is twofold. Firstly, the single photon binary classifier must be constructed such that signal-like photon candidates are given a high score and background-like photon candidates are given a low score. Secondly, cuts on both photon candidate scores must be selected such that the signal efficiency is maximized while the data rate is minimized.

previously used
cuts-based - HLT
currently used

6.2.1 XGBoost-Based BDT Software

XGBoost is a commonly-used software package throughout CMS, as it excels at binary classification problems. Specifically, Boosted Decision Trees (BDTs) are used commonly in CMS software (REF?). BDTs provide a good balance between model complexity and transparency. By tuning a relatively small number of model hyperparameters, as discussed in section 6.2.4, model complexity can be controlled and understood. For the development of this BDT, the python version of XGBoost was used, though the final implementation of the model must be done in the

C-based CMSSW (as discussed in section 6.4).

6.2.2 Signal and Background Samples for Classification

The first – and perhaps most important – step in creating a binary classifier is selecting samples on which to train the BDT. It is important that the signal sample chosen accurately represents the type of events that should be kept at the HLT, while the background sample must encompass all possible source of background events.

Signal Samples

As discussed in (REF?), Monte Carlo simulations exist for various physics processes that contribute to $H \rightarrow \gamma\gamma$. While there are several common production modes for the Higgs boson (as discussed in section (REF?)), gluon fusion Higgs production (ggH) has the largest production cross section. In addition to the large cross section, ggH production has fewer associated productions when compared to VBF or ttH Higgs production. For these reasons, ggH signal MC provides a clean signal that maximizes the photon discrimination ability of a BDT.

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In order to validate the performance of the model, the acceptance of these signal MC events must be quantified. As a result, half of the sample is used for training, while the other half is reserved for later validation. Events are split by event number into two separate categories: even event numbers are used for the training sample, and odd event numbers will be used for the validation of the model performance.

Background Samples

At the trigger level, background simulation poses a major challenge. Background events for $H \rightarrow \gamma\gamma$ fall into three major categories: events containing two photons diphoton, events containing a photon and a jet, and events containing two jets. While it is possible to simulate these background sources, the simulation – especially of the jet-jet events – is challenging and therefore may not perfectly align with the real data. The most obvious resolution to this issue is to simply treat the data sample as background. While, of course, the goal of the trigger is to

Simulated

There are also background events that are not simulated, like

At the trigger level, we are not making the final selection. We only need to reduce the data rate by a large factor, so the data are the best source for our background.

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Table 6.1. Table of input variable names, and a short description of each.

Variable Name	Description
Raw Energy	The total energy deposited in the 5x5 supercluster
R_9	The proportion of the energy contained in the central 3x3 crystals relative to the raw energy
$\sigma_{i\eta i\eta}$	The covariance in terms of crystal spacing (i_η)
η_{SC}	The eta coordinate of the center of the supercluster
η Width	The supercluster width in η
ϕ Width	The supercluster width in ϕ
H/E	The ratio of the total supercluster energy measured in the HCal to that measured in the ECal
ECalPFIso	The isolation energy of the particle deposited in the ECal

preserve signal signatures from the data, it has been shown in chapter ?? that the overall expected count of signal events in this data sample is relatively low. As long as a clean signal sample is provided to the BDT, the incredibly small percentage of possible $H \rightarrow \gamma\gamma$ events in the data should not appreciably mar the model's ability to differentiate signal and background photon candidates. As with the signal MC, even event numbers are used for the training, while odd event numbers are reserved for later validation of model performance.

6.2.3 Determination of Input Variables

After signal and background samples are chosen, input variables for the BDT must be picked to optimize separation chosen such that signal and background events can be separated. The most obvious starting place for this is by analogy to the current HLT; since cuts on the kinematic, isolation, and shower shape variables have been successfully used in the HLT preexisting trigger, a BDT should be able to use these variables to even greater effect. Below, in table 6.1, the chosen input variables are listed along with their definitions.

The chosen variables are largely shower shape and isolation variables, as these can be used to differentiate a legitimate photon signature from a jet shower. A major goal of the BDT is to increase kinematic coverage; relying primarily on shower shape and iso variables ensures that the selection is not biased towards a specific kinematic region. Below are plots of the above input

variables, where the signal MC is shown in black and the data (background) is shown in red.

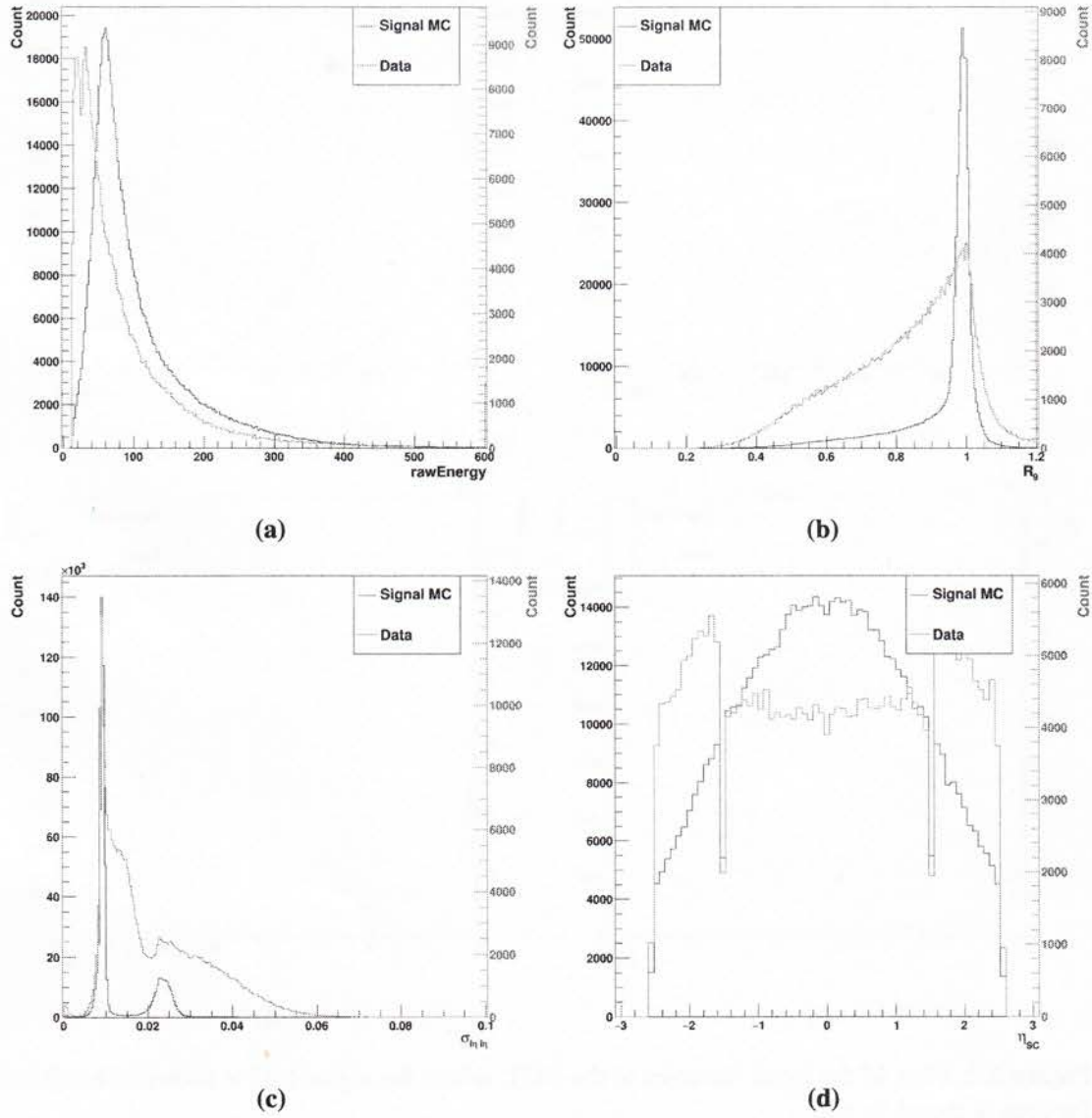


Figure 6.1. Plots of the input variables to the BDT, where the signal MC is shown in black and the data is shown in red.

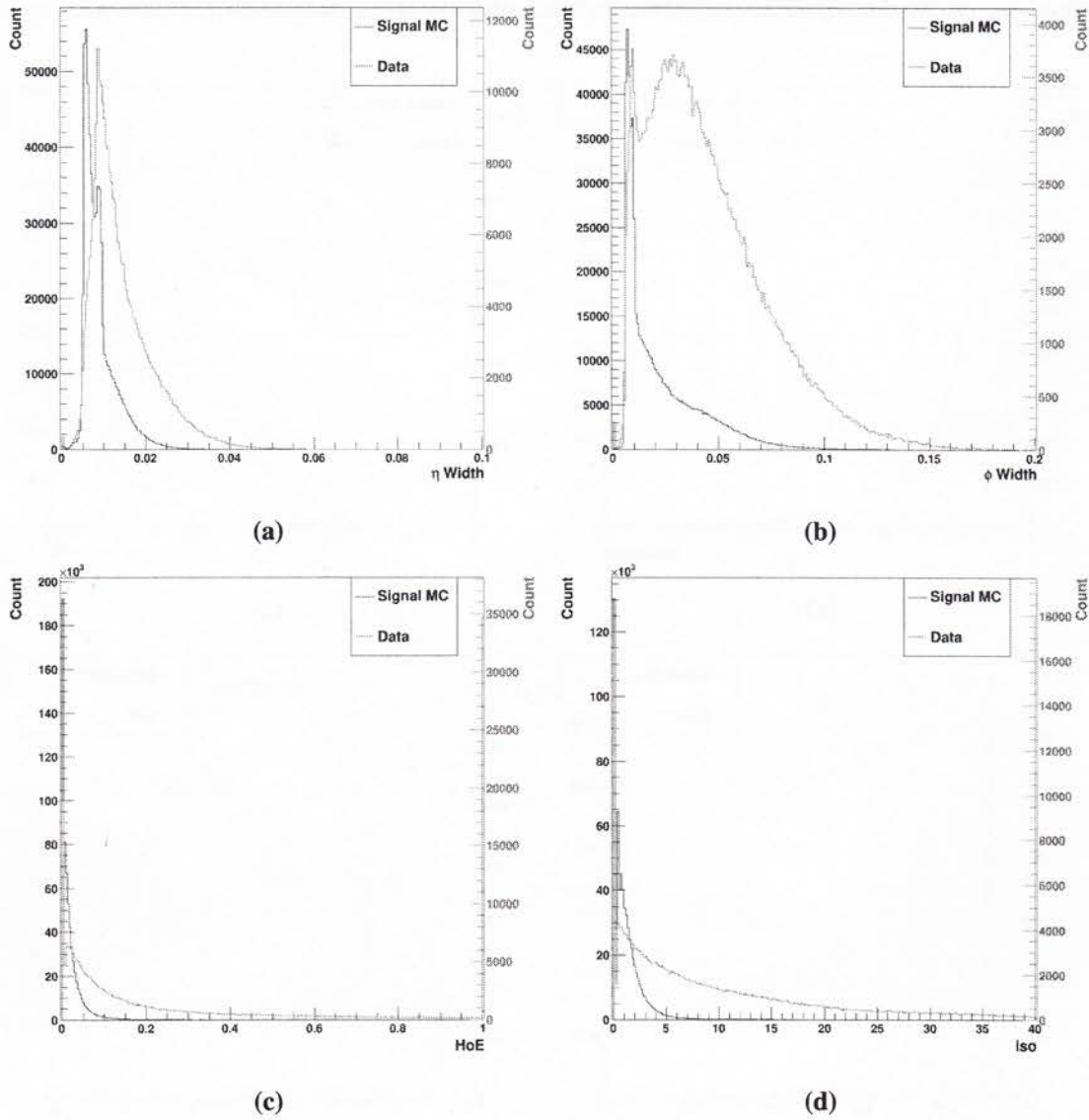


Figure 6.2. Plots of the input variables to the BDT, where the signal MC is shown in black and the data is shown in red.

6.2.4 Hyperparameter Tuning

Once the input samples and variables are determined, the technical details of the model must be optimized. The structure of the BDTs is determined by the hyperparameters of the model as discussed in chapter 5. The most fundamental hyperparameters are the maximum depth, the learning rate, and the total number of trees in the model. The max depth and number of trees

directly determine the complexity of the model. A model with more trees will, based on the boosted gradient descent method discussed in chapter 5, more precisely estimate the function that maps the chosen input variables from the training sample onto the correct labels of signal versus background. To mitigate the risks of overtraining the model, XGBoost early stopping is used to determine the number of trees. As the software adds additional trees to the ensemble, it measures the chosen objective function at each step on a testing sample not used in the training. Early stopping ends the training of the model when the chosen metric hasn't improved in a chosen number of rounds. For this training, 50 early stopping rounds were used and the area under the curve (AUC) was used as the target metric. So, then, this hyperparameter is essentially fixed while the others must be optimized.

The maximum depth controls the complexity of each individual tree in the ensemble. As discussed in chapter 5, the depth determines the number of layers each tree is able to employ. Deeper trees generally allow more precise mapping from input values to output scores at the cost of computation time; each layer of depth constitutes another set of weights that must be computed for each tree. Since it is vital that the computation time is minimized at the trigger level, this is a major consideration. Deeper trees also tend to overfit the training data more, as there are essentially more degrees of freedom with which to fit the mapping function. So, the max depth must be kept as low as possible to avoid both overfitting and unreasonable computation time.

The learning rate controls the size of the individual feature weights from each new tree added to the ensemble. A higher learning rate allows each tree to have higher weights, which will therefore affect the model performance more drastically. Lowering the learning rate makes the model more conservative, as it reduces the overall change to the model predictions provided by each additional tree. Having a higher learning rate generally reduces the number of trees necessary to converge, but decreases the accuracy since statistical fluctuations within each tree can be assigned higher weights.

It was found that the max depth and learning rate are best optimized in tandem; as the



trees become more complex (higher max depth), it was found that making the individual steps more conservative (lower learning rate) led to less overfitting and higher performance. As a result, a 2D grid search was used to optimize these two hyperparameters. A range of both max depth and learning rate was defined, then a model was trained for each combination. The resulting models were appraised in terms of not only their objective scores (AUC and logistic loss in this case, both described in chapter 5), but also in terms of the number of trees and the overall model size in megabytes. Since the model must be loaded into RAM at the trigger farm and then the weights applied to each photon in each incoming data event, a model taking up more space will increase both the computation time and memory load at the trigger level.

After many trials, the best balance of model performance, minimization of overfitting, and model size was found to be a max depth of 7 and a learning rate of 0.25. Experimentally, it was determined that many of the other hyperparameters had minimal impact on the model performance and size, so they were therefore left at their default values.  

6.3 Determination of BDT Cuts

After the tuning and training of the model was optimized, a score is generated for each photon in each event. Each event may contain several photon candidates, but the job of the HLT is to determine if any pair of candidates exists that plausibly originates from $H \rightarrow \gamma\gamma$. Given that the BDT was trained such that photons originating from $H \rightarrow \gamma\gamma$ (MC) are assigned a high score and background processes are given a low score, the photon candidates are first sorted by the output BDT score. Then, the highest two scores can be selected and the values analyzed for both signal MC and data events.

Since the HLT path simply renders a decision about whether an event contains a $H \rightarrow \gamma\gamma$ process, cuts on the pairs of output scores must be chosen such that the number of signal events kept is maximized (high signal efficiency) while the total number of data events kept is minimized (low trigger rate). 

Below, in figure 6.3, the high and low scores for signal MC are shown at left. Since these are purely signal events, it is expected that the scores to be clustered near 1. As can be seen, the higher of the two scores (along the x axis) tends to be above 0.9 for a large percentage of events. Meanwhile, while the lower of the two scores does cluster above 0.8, there is still a large band of score values extending down to 0. This means that there are many signal events for which one photon has a very high score, but the second photon has a lower score. ^{-candidate}

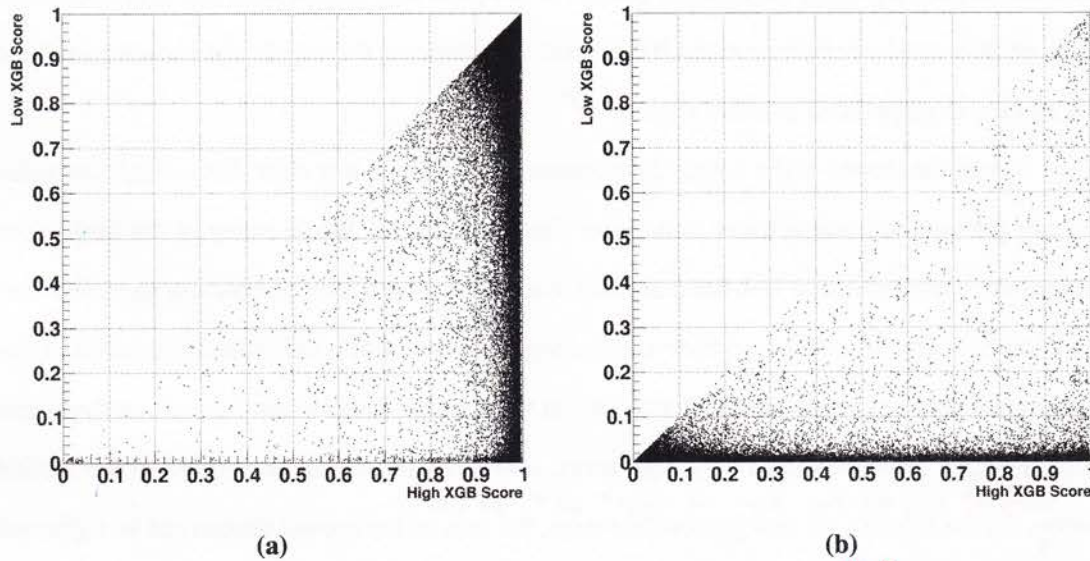


Figure 6.3. XGBoost output scores for two selected photon candidates in the Winter 2024 ggH signal MC. The higher of the two scores is plotted on the X-axis, while the lower of the two is plotted on the Y-axis. ^(a)

Next, consider the scores of the data sample shown at right in in 6.3. Given the rarity of $H \rightarrow \gamma\gamma$ events at the trigger level, it is expected that essentially all of these data events correspond to background processes. As a result, the vast majority of these data photon candidates should be assigned a BDT score close to zero. As can be seen, the lower of the two photon scores very rarely extends above 0.1. Though the higher of the two scores is often close to 0, there are still many events where the value extends towards 1. Since this BDT is designed to identify individual photon candidates, it can be inferred that many of these data events have one real photon (which is a major source of background as discussed in section 6.2.2). ^{or more}

As discussed in section 6.2.1, the single photon BDTs are trained separately for photon candidates in the barrel and those in the endcap. Due to the separate trainings – as well as the differences in kinematics and detector geometry – the score distributions for barrel and endcap photon candidates are not perfectly equivalent. As a result, the chosen cuts ~~must be~~ ^{are} tuned separately for the barrel and endcap photons. This presents an issue, as diphoton events may contain two barrel photon candidates (BB), two endcap photon candidates (EE), or one photon candidate in each region of the detector (BE). The simplest way to deal with this is to choose the single photon cuts by reference to the BB and EE distributions, then apply the chosen barrel and endcap cuts to all photons in either region. ✓

Given the shapes of the output distributions in figure 6.3, it is clear that a single cut value for both photons will not achieve maximum signal efficiency. While many of the high score photons are clustered above 0.9, there are still many events with a second photon candidate that has a lower score. In order to preserve these signal events while still removing much of the background, cuts can be selected such that the cut value on the lower score depends on the higher score. For first photons with a very high score, a very loose cut can be applied to the second photon. ^{without increasing the trigger rate much.} As the score of the first photon decreases, the cuts on the second photon can be tightened. In the case where the first photon has a relatively low score, a tighter cut is applied on the second photon to help reduce the number of events with two bad photon candidates in the data while still preserving much of the signal. Since the data at the trigger level is strongly dominated by di-jet or single photon events, these cuts can drastically reduce the data rate while still preserving much of the signal. These cuts were experimentally determined, and are tabulated in table 6.2.

Table 6.2. Chosen score cuts for BB and EE events

(a) Score cuts for BB events

High Score	Minimum Low Score
0.92	0.02
0.85	0.04
0.30	0.14

(b) Score cuts for EE events

High Score	Minimum Low Score
0.95	0.04
0.85	0.08
0.5	0.20

Events that satisfy these requirements can be plotted on top of figures 6.3. In figure 6.4 and 6.5 below, the selected events for BB and EE can be seen, respectively, in red. Yellow lines are added to the plots to demarcate the cuts. These plots contain both the signal MC (left) and the Data (right).

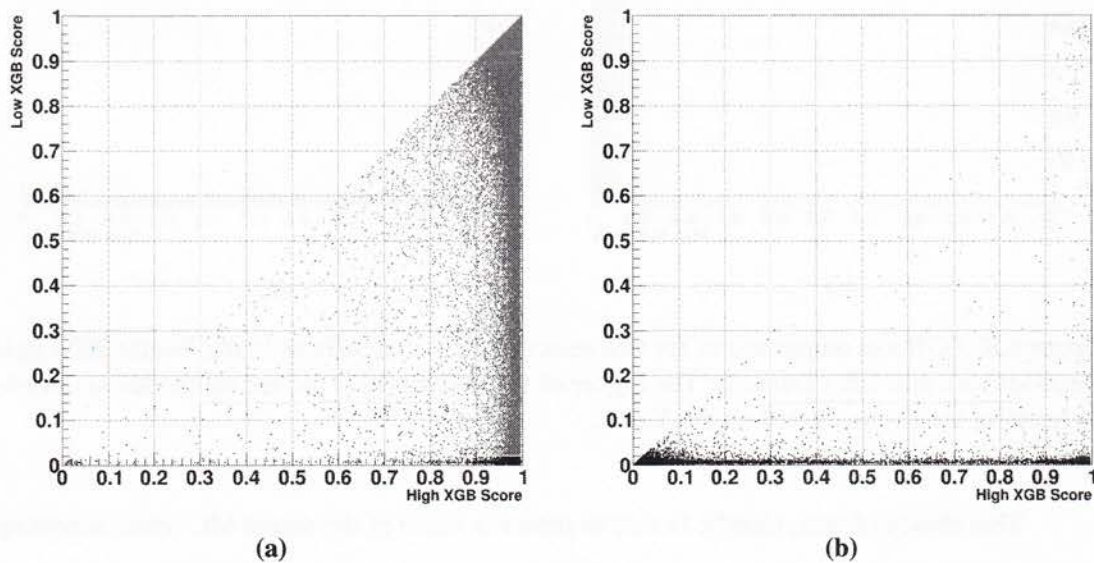


Figure 6.4. XGBoost output scores for two selected photon candidates in the Winter 2024 ggH signal MC (a), and 2023 Data (b). The higher of the two scores is plotted on the X-axis, while the lower of the two is plotted on the Y-axis.

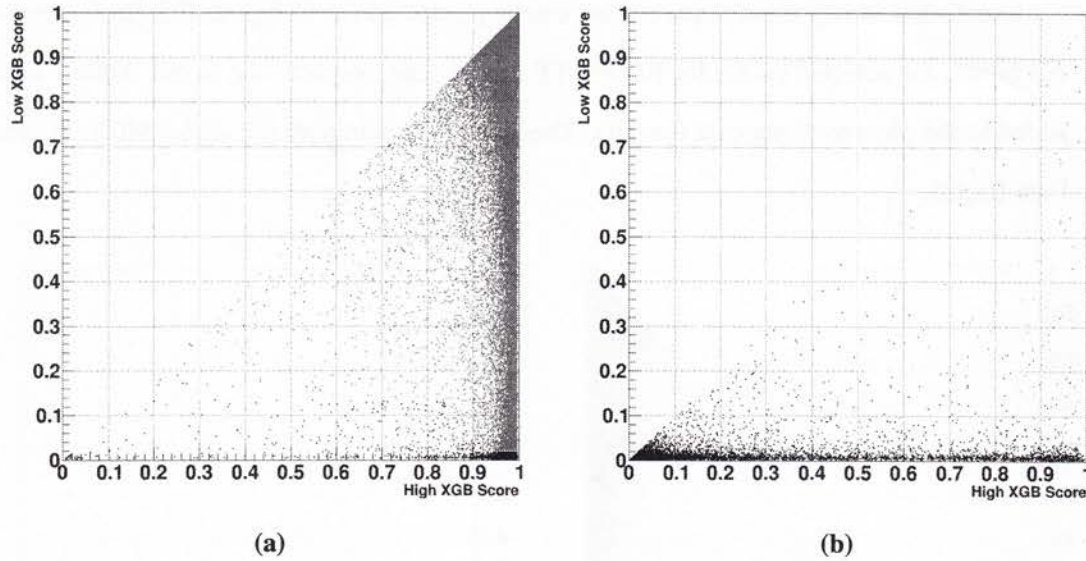
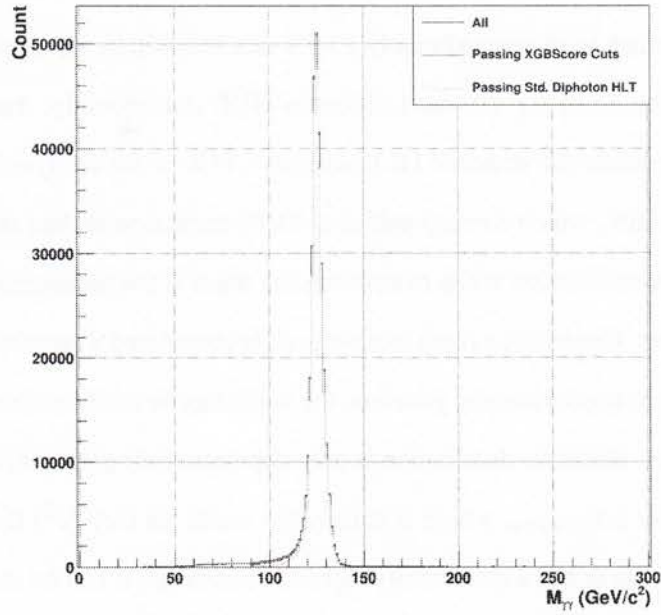


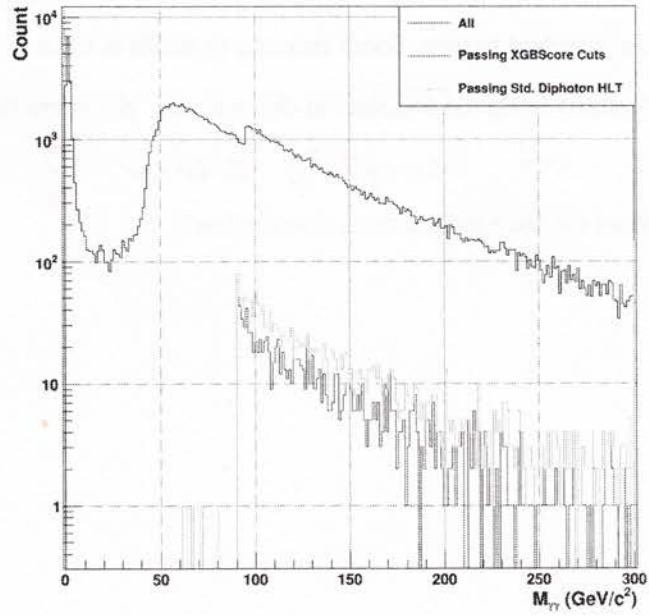
Figure 6.5. XGBoost output scores for two selected photon candidates in the Winter 2024 ggH signal MC (a), and 2023 Data (b). The higher of the two scores is plotted on the X-axis, while the lower of the two is plotted on the Y-axis.

This choice of cuts, clearly, is able to preserve much of the signal MC while accepting very few of the data events examined. While the overall signal efficiency and data rate are key metrics to appraise our MVA-HLT, it is also important to examine where in the kinematic phase space events are being gained relative to the standard diphoton HLT. While a more complete measurement of this will be shown in chapter 8, a preliminary analysis can be made with the cuts applied in this initial development framework.

Below, plots of selected kinematic variables are shown. The events accepted by the current diphoton HLT are shown in green, while the events accepted by the new MVA-HLT are shown in blue. As with the previous score plots, the Winter24 GGH signal MC will be shown at left, whereas the 2023 Data will be shown at right.



(a)



(b)

Figure 6.6. Diphoton mass for the signal MC (a) and data (b). Events accepted by the MVA-HLT are shown in blue, while events accepted by the standard HLT are shown in green.

Beginning with the mass, it can immediately be seen that MVA-HLT accepts 94.2% of signal MC events, which is an approximately 13.6% relative gain in signal efficiency compared to the 82.9% acceptance of the standard diphoton HLT. Additionally, the data rate appears considerably lower; while the standard HLT accepts 1.44% of the analyzed events, the MVA-HLT only accepts 0.86%, which corresponds to a 40.3% reduction in data rate.

One major consideration when examining the mass is the measured mass width of the accepted signal events. Generally, a more narrow peak is considered a cleaner signal. By treating the mass distribution as approximately gaussian, the width can be easily understood by measuring the standard deviation. Since the distribution is only approximately gaussian, it is safer to instead construct the quantity $2\sigma_{effective}$, which is simply the width (in GeV/c^2) that contains 95% of the peak. Although this is not a statistically rigorous construct, it can be used to compare the approximate shapes of similar distributions in an internally consistent way. It can be seen that the $2\sigma_{effective}$ is marginally increased for the MVA-HLT, reaching 6.58 compared to the 6.07 of the events accepted by the standard trigger. Some increase in width is to be expected, as accepting more events will necessarily bring the distribution closer to the "all" curve that has a wider width than either selection.

and accepting lower P_T photons...

(Statements about the background mass distribution?)

~~Fig 6.7~~

in Fig 6.7

Above, the next kinematic variable shown is the η_{sc} of both photons plotted together. As seen in both signal and background, the separation between the ECal Barrel (EB) and ECal Endcap (EE) can be seen at $\eta_{sc} \approx \pm 1.5$. For the signal, throughout the barrel, both the MVA-HLT and ~~standard~~ HLT accept photon candidates in relatively constant proportion to the total distribution, though the MVA-HLT has overall improved acceptance. While the MVA-HLT still has relatively constant acceptance in the EE, it can immediately be seen that the standard HLT is excluding many events in the lowest- η_{sc} region of the endcap. Investigations have shown that this is due to the R_9 cuts made within the trigger. This represents a reduction in kinematic coverage for the standard HLT, whereas the MVA-HLT is able to resolve this issue and maintain maximum kinematic coverage.

Considering the background, it can immediately be seen that the MVA-HLT has a relatively flat acceptance in both the EB and EE, while the standard HLT appears to accept more data at the extreme- η_{sc} regions of the endcap. As can be seen from the MC, this region tends to contain less of the $H \rightarrow \gamma\gamma$ signal events and therefore represents a lower signal to background ratio.

6.8

Finally, consider the transverse energy of the photon candidates, E_T . Here, both candidates are again shown on the same plot, which explains the double-peaking behavior in the background distribution. The standard diphoton HLT contains cuts at 30 and 22 GeV/c^2 for the lead and sublead photon, respectively. The sublead cut can clearly be seen in both the signal, at left, and the background at right. While the new MVA-HLT selection accepts more of these low- E_T photons, it is clear that little of the gain comes from this kinematic region. Below about 30 GeV/c^2 , both selections tend to cut out many photons from the total distribution. In the case of the MVA-HLT, this is likely due to those photon candidates being of lower quality and therefore being assigned a lower BDT score.

Below, in Table 6.3, the acceptance percentages are given for the 3 different diphoton η_{sc} configurations, as well as the total acceptance for both signal and background, compared to the standard HLT cuts.

Table 6.3. Signal efficiencies and background acceptances for both triggers, broken down by diphoton η_{sc}

(a) Signal Efficiencies			(b) Data Acceptances		
Region	MVA-HLT(%)	Std. HLT(%)	Region	MVA-HLT(%)	Std. HLT(%)
B-B	95.93	89.32	B-B	0.65	1.06
B-E	91.91	74.10	B-E	1.20	2.17
E-E	89.46	76.85	E-E	0.58	1.07
All	93.86	82.64	All	0.88	1.56

6.4 Final Implementation in HLT Menu

As discussed in section 3.3.3, triggering and event reconstruction is handled by the distributed CMSSW software framework. As a result, any proposed trigger developments must be deployed in a central CMSSW release. While the software strategy is the same as the original python implementation, the code must of course be rewritten in C and adapted to work within the trigger.

As described in section 3.3, the HLT operates on all photon candidates in an event, then

simply renders an event-by-event decision if any pair of photons satisfy the chosen conditions. As a result, the first step of the implementation is to create a module that calculates the BDT score for each photon candidate in every event. Once these scores are stored within the trigger path, another module must be constructed to check if any possible pair of photon candidates passes the two-fold cuts described above. Importantly, due to the constraints on storage space, the scores of the individual photon candidates cannot be saved to the event. As a result, the only output from the MVA-HLT is a decision on whether the event should be accepted or rejected.

Luckily, these two modules are the only software additions required to add the BDT to the trigger menu. Beyond that, standard modules can be used (with some modifications to the cuts made) within ConfDB to construct the entire trigger path as discussed in section ?? . The first step of the HLT path is to require the necessary L1 triggers. The standard L1 suite required by the ~~existing~~ HLT is used here, which consists of a combination of single and double EGamma triggers as described in section 3.1.

In order to reduce the amount of MVA scores ~~that must~~^{to} be calculated, the very loose kinematic cuts are applied to all photon candidates in the event as the first step. While these cuts are quite loose, they still serve to reduce the number of candidates considered. Since many data events may have many photon candidates, this will help reduce the time taken to calculate the BDT scores. (Is this true? Do individual photon candidates get removed at the HLT level, or just events?)

With the preliminary cuts in place, the module that calculates the scores is placed in the HLT path, followed by the module that makes the cuts as described in section . Finally, a combined module is created that combines these BDT cuts on both photons with a mass cut of $M_{\gamma\gamma} > 90 \text{ GeV}/c^2$. So, then, the whole HLT path serves to check if there is any possible pair of photon candidates that passes the loose kinematic cuts, the pairwise BDT cuts, and the $M_{\gamma\gamma}$ cut. If any pair satisfies these conditions, a flag is added to the event and all photon candidates are kept at the RAW level.